

Multiple Choice (10 marks)

1. Which of the following is the joint probability of H, U, P, and W described by the given Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$? [note: as the product of the conditional probabilities]
 - (a) $P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)$
 - (b) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)$
 - (c) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$
 - (d) None of the above
2. The numerical output of a sigmoid node in a neural network:
 - (a) Is unbounded, encompassing all real numbers.
 - (b) Is unbounded, encompassing all integers.
 - (c) Is bounded between 0 and 1.
 - (d) Is bounded between -1 and 1.
3. Adding more basis functions in a linear model, pick the most probably option:
 - (a) Decreases model bias
 - (b) Decreases estimation bias
 - (c) Decreases variance
 - (d) Doesn't affect bias and variance
4. Which of the following sentence is FALSE regarding regression?
 - (a) It relates inputs to outputs.
 - (b) It is used for prediction.
 - (c) It may be used for interpretation.
 - (d) It discovers causal relationships
5. Which image data augmentation is most common for natural images?
 - (a) random crop and horizontal flip
 - (b) random crop and vertical flip
 - (c) posterization
 - (d) dithering

True / False (10 marks)

Answer True or False

6. True or False: RoBERTa pretrains on a corpus that is approximately 10 times larger than the corpus BERT was pretrained on.
7. True or False: The sigmoid function is a commonly used nonlinear activation function in neural networks.
8. True or False: Fully connected networks are the best choice for classifying high-resolution images due to their ability to capture spatial hierarchies.
9. True or False: Both RELUs and sigmoids are monotonic activation functions used in neural networks.
10. True or False: Stuart Russell is a prominent professor known for his work on existential risks associated with AI.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function that matches a string that has an a followed by one or more b's.
12. Write a function to count the number of elements in a list which are within a specific range.

End of Paper